Riccardo Pala

Software Engineer | Java, Concurrency & Distributed Systems

Software Engineer with 3+ years building production Java/Spring Boot systems for web and mobile banking. Currently focused on multithreaded processing, concurrency and low-latency performance optimization in distributed services. Experienced with Kafka, cloud-native delivery and technical ownership, with additional expertise in applied AI and RAG infrastructure.

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Professional Experience

Software Engineer — PFM & Performance
Jun 2026 - Present

Silicon Shoring FS Reply • Banca Widiba • Milan

- **Concurrency & Low Latency:** Work on multithreaded backend processing and low-latency execution paths for a personal finance management platform, with a focus on performance and responsiveness.
- **Performance Engineering:** Investigate processing bottlenecks and optimize transaction workflows to improve efficiency and reliability.
- **Backend Development:** Build Java/Spring services for transaction categorization, merchant normalization, geolocation and recurring-payment detection.
- **RAG Infrastructure:** Develop backend and PostgreSQL/pgvector retrieval components for an LLM-powered feature; coordinate integration with the Python/LLM engineer.

Software Engineer & Technical Lead
Jun 2023 - Present

Silicon Shoring FS Reply • BPER Banca • Milan

- **Production Systems:** Design and deliver Java/Spring Boot REST APIs and microservices for web and mobile banking, emphasizing maintainability, scalability and security.
- **Distributed Architecture:** Develop Kafka-based event-driven integrations and Oracle data services; troubleshoot production incidents and performance issues.
- **Technical Ownership:** Contribute to system design, code reviews and engineering decisions, supporting colleagues and coordinating delivery across teams.

Education

M.Sc. Computer Science & Engineering (AI)
2020 - 2023

Politecnico di Milano

Final Grade: **110/110 cum laude**.
Specialization: Deep Learning, Machine Learning, Recommender Systems, Game Theory.

Research Exchange (XAI)
2022 - 2023

Universidad Politécnica de Madrid

Thesis: *Robustness of Ensemble Explanations (AXOM)* (Executive Summary).
Rated **10/10 (Top 5% of best thesis work)**.

Projects & Publications

Product Engineering (Clup & MuSeek)
Built end-to-end web and native Android products with J2E/Tomcat and Kotlin, including service-layer algorithms and API-backed application flows.

ACM RecSys 2022 Conference Publication
Co-authored *"Lightweight Model for Session-Based Recommender Systems with Seasonality Information in the Fashion Domain"*. Explored neural-network and matrix-factorization approaches for session-based recommendation in Python.

eCommerce Online Learning System
Designed 6 online learning algorithms (UCB, Thompson Sampling) for behavior-based intelligent pricing.

TECH STACK

Backend & Performance

Java 21 | Spring Boot | Concurrency | Multithreading | Low-Latency Processing | Performance Optimization | REST APIs | Microservices | System Design

Distributed Systems & Delivery

Apache Kafka | Oracle SQL | PostgreSQL / pgvector | Event-Driven Systems | OpenShift / Kubernetes | Docker | AWS | Jenkins / CI/CD | Dynatrace

Applied AI & Algorithms

RAG Systems | Vector Retrieval | Python | PyTorch | Recommender Systems

CERTIFICATIONS

Confluent Certified Developer (CCDAK)
Confluent Certified Developer for Apache Kafka (Issued: Feb 2026)

AWS Certified AI Practitioner
IN PROGRESS
Expected: Q4 2026

LANGUAGES

English — Fluent (C1, TOEIC 955)
Spanish — Conversational (improving)
Italian — Native